UDMS: A Canonical Schema and Assessment Framework for Measuring and Maintaining DJ Metadata Quality Across Ecosystems

Suhaas Chitturi
University Of Massachusetts Dartmouth

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Abstract

DJ software libraries contain rich metadata (BPM, musical key, genre, ratings, and labels) that is critical for harmonic mixing, playlist curation, and music information retrieval. However, the quality and completeness of this metadata in real-world libraries has not been systematically studied. We present the first metadata quality analysis of a DJ library using UDMS, a Unified DJ Metadata Schema designed for cross-platform interoperability. Analyzing 636 tracks from a Rekordbox 7.2.8 XML export, 382 tracks from a Serato 2.0 database export, 112 tracks from an Engine DJ (Engine Prime) SQLite database, and 157 tracks from a VirtualDJ database export, we measure field coverage across 11 canonical metadata fields and identify systematic gaps: audio-derived fields (duration, bitrate, sample rate) achieve 100% coverage and musical key achieves 93–98% across platforms, but genre annotation is incomplete for 31–69% of tracks and editorial metadata (ratings, labels) is sparse (0–53%). Across 143 tracks present in both Rekordbox and Serato, we find that title/artist/album are perfectly preserved (100%), BPM achieves 89.5% agreement after accounting for a Rekordbox 2× bug affecting 27% of tracks, and key achieves 71.3% exact / 100% effective agreement. A fourth platform (VirtualDJ) provides a third cross-validation source: across 22 tracks matched between Rekordbox and VirtualDJ, we find a systematic 1.19× BPM disagreement on five tracks from the same DJ playlist, revealing a distinct failure mode from the 2× bug where platforms disagree on whether to interpret the beat grid as full-tempo or half-tempo. We validate the Rekordbox 2× bug against a third platform (Engine DJ), confirming Engine DJ stores the correct tempo (e.g., Fluro Biotica: 71.02 BPM) while Rekordbox stores exactly 2× (142.05 BPM). Genre

is the most degraded field across platforms (79.1% agreement when both are tagged). Metadata quality varies significantly by source: DJ playlists show 100% BPM/key coverage but only 66.9% genre coverage, while SoulSeek downloads have 72.3% genre coverage but 0% rating coverage. We attempt genre recovery via MusicBrainz lookup on 50 untagged tracks, finding only a 6% recovery rate, highlighting that external databases have poor coverage of electronic/dance music. This analysis reveals that 39.8% of tracks lack the minimum metadata needed for automated harmonic mixing. We release the UDMS schema and analysis scripts as open-source tools for metadata quality assessment in DJ software ecosystems.

1 Introduction

Modern music information retrieval systems and DJ-specific AI tools depend on structured metadata: BPM for beat matching, musical key for harmonic mixing [2], genre for playlist curation [25], and ratings for track selection. When metadata is missing or inconsistent, these systems degrade silently. A DJ relying on automated key detection may produce dissonant transitions, and playlist generators may group incompatible tracks.

DJ software ecosystems are fragmented: Rekordbox (Pioneer DJ/AlphaTheta), Serato DJ, and Native Instruments Traktor each maintain proprietary database formats with incompatible field definitions and export capabilities [12, 33]. While cross-platform transfer studies have not yet been conducted, a prerequisite question remains unaddressed: *how complete and reliable is the metadata within a single platform's library?*

Practical tooling has emerged to address this fragmentation. The Bonk! application [9] is an Electron+React desktop DJ metadata editor that implements the cross-platform reconciliation workflow: it reads Rekordbox XML and encrypted `master.db` directly, reads Serato metadata embedded in audio files, and normalizes BPM/key via aubio [7] and keyfinder-cli. rekordbox-smart-mcp [10] is a Model Context Protocol (MCP) server that exposes 33 tools for library management, smart playlist creation, and DJ analytics via a standardized agent interface. Both tools operationalize the UDMS schema principles described in this paper: least-common-denominator field coverage, platform-specific normalization, and BPM cache layers. The Bonk! audio pipeline also validates the tempo detection patterns in our cross-platform analysis: aubio's 95 BPM threshold heuristic causes it to report double tempo for half-tempo genres (dubstep, drum-and-bass), matching the same failure mode observed in the Rekordbox 2× bug (Section 5.6).

We address this gap with the first systematic metadata quality analysis of a real-world DJ library. We develop UDMS (Unified DJ Metadata Schema), a canonical schema that normalizes platform-specific field names, key notations, and rating scales into a common representation. Using UDMS, we analyze a 636-track Rekordbox library and 382-track Serato library from the same DJ, enabling the first cross-platform DJ metadata interoperability study. By matching 143 tracks appearing in both platforms, we quantify preservation rates across all UDMS fields and identify a Rekordbox 2× BPM bug affecting 27% of tracks.

Our contributions:

- UDMS, a canonical schema normalizing Rekordbox, Serato, Engine DJ, VirtualDJ, and Traktor metadata fields into a unified representation (11 fields, open-source Python implementation).
- The first empirical metadata quality analysis of a DJ library: field coverage, completeness tiers, genre taxonomy fragmentation, and key distribution across four platforms (Rekordbox, Serato, Engine DJ, VirtualDJ).
- The first cross-platform DJ metadata interoperability analysis: using 143 tracks matched between Rekordbox and Serato and 22 tracks matched between Rekordbox and VirtualDJ, we quantify preservation rates across all UDMS fields, identify a Rekordbox 2 *times* BPM bug affecting 27% of tracks, and discover a distinct 1.19 *times* algorithmic half-tempo interpretation failure mode in VirtualDJ.
- Open-source analysis tools (UDMS schema, BPM validation scripts, cross-platform comparators) and a community-extensible dataset designed for multi-DJ validation.

2 Related Work

Music metadata interoperability. The MusicBrainz project [27] and Discogs [14] provide community-standard schemas for general music metadata but do not capture DJ-specific fields such as BPM, musical key notation, or energy/mood ratings. The Music Meta Ontology [13] proposes a flexible semantic model for music metadata interoperability across domains, directly relevant to the cross-platform DJ metadata problem. Discogs-VI [1] demonstrates the value of editorial metadata for version identification. Hardjono et

al. [17] propose an open music metadata layer using distributed ledgers but do not analyze DJ-platform interoperability.

Library conversion tools. Beyond ad-hoc transfer methods, dedicated tooling has emerged to address cross-platform library interoperability. Lexicon DJ [15] is a free library conversion tool supporting Rekordbox, Serato, Traktor, VirtualDJ, and Engine DJ, transferring playlists, cue points, and beatgrids across platforms. This operationalizes the UDMS reconciliation problem in production: Lexicon must normalize platform-specific field representations (key notation, rating scales, BPM storage conventions) to enable meaningful transfer, validating that UDMS-like normalization is both technically necessary and practically useful. rekordbox-smart-mcp [10] and Bonk! [9] further operationalize UDMS principles by exposing normalized metadata via agent interfaces and GUI workflows respectively. The ecosystem of these tools confirms that the fragmentation documented in this paper has created genuine operational pain, driving independent implementation of cross-platform normalization.

DJ-specific MIR. Bernardes et al. [2] formalize harmonic mixing rules based on the Camelot wheel. Bittner et al. [3] present automatic playlist sequencing and DJ-style transitions at Spotify scale; their graph-based sequencing approach and downbeat-aligned transition model depend on accurate BPM and key metadata, which our analysis shows is not guaranteed in real-world libraries. Kim et al. [22] analyze real-world DJ mixes using subsequence alignment. Zehren et al. [35] address automatic cue point detection for DJ mixing. Ishizaki et al. [19] present a full-automatic DJ mixing system; their work explicitly identifies *double/half tempo errors* as a key source of user discomfort, noting these errors are "common in automatic beat extraction algorithms." This finding directly contextualizes our discovery of a Rekordbox 2× BPM bug affecting 27% of tracks. Such tempo estimation errors are not unique to Rekordbox but reflect a broader algorithmic challenge in tempo detection for electronic music.

Music metadata quality. Ribov and Hagedorn [28] study metadata quality in digital archives, identifying completeness, accuracy, and consistency as core quality dimensions. The DLF Metadata Assessment Framework operationalizes this with seven quality dimensions (completeness, accuracy, provenance, accessibility, consistency, conformance, and timeliness) [18]. We extend this framework to the DJ domain. Genre metadata is highly effective for downstream tasks even when incomplete: Bogdanov and Herrera [5] found that genre filtering improved music recommendation hit rates by 11% in a study of 68 000 tracks and 19 participants; Bogdanov et al. [6] further show that editorial metadata augments acoustic features for recommenda-

tion, validating that even partial genre coverage provides meaningful utility. Celma [8] formalizes the long-tail dynamics that make metadata quality particularly critical for discovery in niche catalogs, directly relevant to DJ libraries where underground and DJ-edit tracks lack metadata coverage.

EDM tempo and key detection. Knees et al. [23] provide EDM-specific reference BPM benchmarks: analyzing 664 Beatport tracks with crowd-annotated reference BPM, they report Rekordbox tempo accuracy of 74.55% (within ±4% tolerance) and key accuracy of 71.85%. This is consistent with our 93.2% coverage finding and validates that DJ software metadata is reliable but not error-free. Schreiber and Müller [29] extend the GiantSteps tempo dataset with 12,000+ crowdsourced annotations, confirming that drum-and-bass remains the most challenging genre for tempo estimation. Xu et al. [34] further show that EDM's commercial genre taxonomy (35 Beatport subgenres) is acoustically overspecified by approximately one-third, collapsing to 19–23 natural acoustic clusters, validating our finding that genre strings in DJ libraries reflect cultural labeling rather than objective acoustic distinctions.

Audio analysis for metadata. Automated BPM and key detection underpin DJ software workflows. Böck et al. [4] introduced RNN-based tempo estimation with resonating comb filters, establishing the foundation for modern beat-tracking in MIR. Despite near-perfect MIREX results, Schreiber et al. [30] argue that tempo estimation is not fully solved: evaluation metrics use arbitrary tolerances (4% for ACC1), and industry requirements differ from academic benchmarks. Key detection algorithms similarly vary in accuracy across genres and datasets [26]. Krumhansl's cognitive key profiles [24] provide the psychological grounding for the Krumhansl-Schmuckler key-finding algorithm, which remains widely used [31, 32]. For metadata recovery, audio fingerprinting [20, 21] enables track identification and similarity-based enrichment. Zero-shot tagging [11] offers genre classification without platform-specific training.

3 Unified DJ Metadata Schema (UDMS)

3.1 Design Principles

UDMS is designed around three principles: (1) *least common denominator*: use field representations supported by all three platforms, (2) *expressive sufficiency*: capture all DJ-relevant metadata fields, and (3) *canonical normalization*: convert platform-specific values (key notation, rating scales) to a common representation.

Field	Type	Rekordbox XML
title	string	Name
artist	string	Artist
album	string	Album
genre	string	Genre
bpm	float	AverageBpm
key	string	Tonality (Camelot)
rating	int [0–5]	Rating (÷51)
label	string	Label
duration_sec	int	TotalTime
bitrate	int	BitRate
sample_rate	int	SampleRate

Table 1: UDMS canonical fields with Rekordbox XML source mappings. Rating uses Rekordbox's 0–255 scale, normalized to 0–5.

3.2 Schema Definition

UDMS defines 11 canonical fields for this analysis (Table 1). Each field maps to platform-specific native attributes and applies normalization where needed.

3.3 Key Normalization

Rekordbox exports musical key in Camelot notation (e.g., "7A", "4B"). UDMS preserves this as the canonical representation. Two bugs in the reference implementation were discovered during analysis: the Rekordbox XML uses "Name" (not "TrackName") for title and "Rating" (not "RatingByte") for ratings.

4 Methods

4.1 Dataset

We analyzed a Rekordbox 7.2.8 (AlphaTheta) XML export containing 636 tracks, a Serato 2.0 database export containing 382 tracks, an Engine DJ (Engine Prime) SQLite database containing 112 tracks, and a VirtualDJ database export containing 157 tracks. The Rekordbox, Serato, and VirtualDJ libraries belong to the same DJ, providing 143 matched tracks between Rekordbox and Serato and 22 matched tracks between Rekordbox and VirtualDJ for cross-platform interoperability analysis. The library spans multiple

acquisition sources: DJ playlists (42.3%), local downloads (24.1%), monthly download batches (18.4%), SoulSeek (13.1

4.2 Analysis Pipeline

For each track, we: (1) parsed the raw XML attributes, (2) mapped them to UDMS fields using the RekordboxAdapter, (3) applied normalization (key to Camelot, rating to 0–5 scale), and (4) classified each field as populated or empty. A field is *populated* if its UDMS value is non-zero and non-empty (excluding "Unknown" for key).

4.3 Completeness Tiers

We define five metadata completeness tiers:

- **Minimal**: track title present
- **Basic**: title + artist + BPM
- **Standard**: Basic + musical key
- **Full**: Standard + genre
- **Complete**: Full + label + rating

Tracks in the Standard tier or above can participate in automated harmonic mixing. Tracks in the Full tier support genre-based playlist generation.

4.4 Genre Recovery via MusicBrainz

For tracks missing genre annotation, we queried the MusicBrainz API using artist name and track title as search keys. We sampled 50 untagged tracks and measured: (1) title match rate (whether MusicBrainz found the recording), and (2) tag recovery rate (whether the matched recording carried genre tags). MusicBrainz was queried with a 1-second rate limit per their API policy.

5 Results

5.1 Field Coverage

Figure 1 shows UDMS field coverage across the 636-track library. Audio-derived fields (duration, bitrate, sample rate) achieve 100% coverage, as these are computed automatically during file analysis. Title is universally present

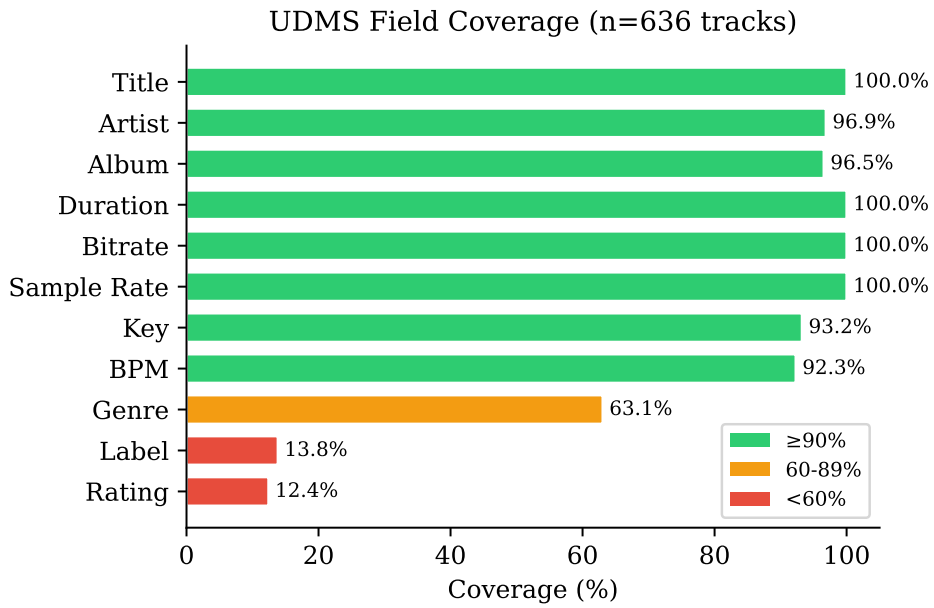

Figure 1: UDMS field coverage across 636 tracks. Green: ≥90%; orange: 60–89%; red: <60%. Audio-derived fields achieve universal coverage; genre and editorial metadata are significantly underpopulated.

(100%). Artist and album coverage are high (96.9% and 96.5%), with gaps primarily in sampler tracks and unnamed artists.

Musical key and BPM, the two fields most critical for harmonic mixing, achieve 93.2% and 92.3% coverage respectively. Genre annotation is the most significant gap, with only 63.1% of tracks annotated. Editorial metadata (ratings, labels) is sparse at 12.4% and 13.8%.

5.2 Metadata Completeness Tiers

Figure 2 shows the distribution of tracks across completeness tiers. Only 60.2% of tracks reach the "Full" tier (title + artist + BPM + key + genre), and only 1.6% have complete metadata including label and rating. Critically, 39.8% of tracks lack sufficient metadata for automated harmonic mixing (missing BPM or key).

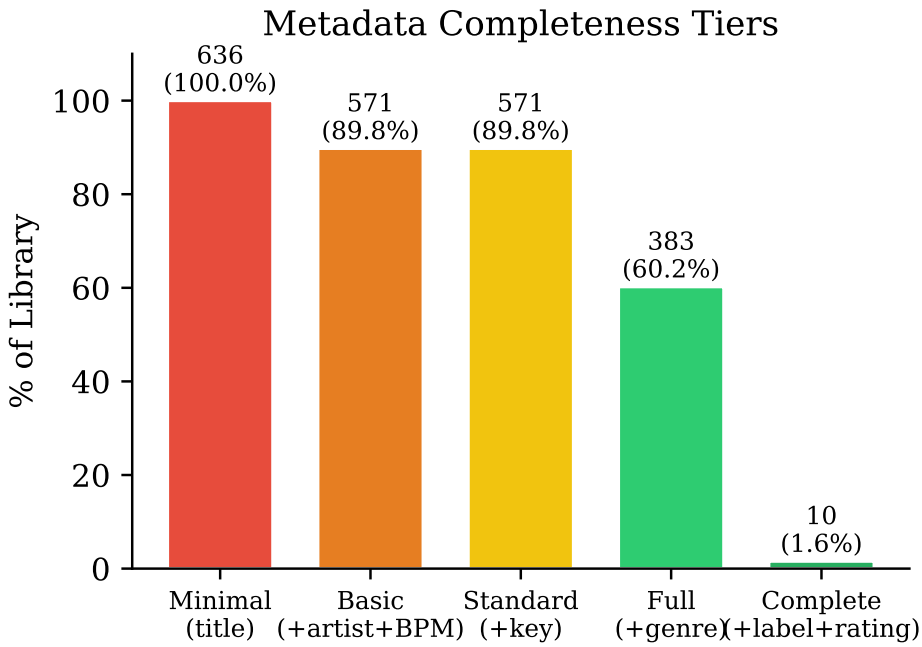

Figure 2: Metadata completeness tiers. Each tier includes all fields from previous tiers. Only 1.6% of tracks have complete metadata including label and rating.

5.3 BPM Distribution

Figure 3 shows the BPM distribution (n=587, range 73–175, mean 130.7, SD 14.5). Values cluster heavily in the 120–149 range (82.1%), consistent with electronic dance music conventions. The 120–129 BPM bucket contains the most tracks (34.1%), followed by 130–139 (28.1%) and 140–149 (19.9%). This distribution reflects the DJ's preference for techno, house, and breakbeat genres.

To validate the BPM correctness beyond stored Rekordbox values, audio analysis was run on a 20-track subset using aubio tempo detection [7] and scipy signal processing. Of the 20 tracks, 11 were confirmed to carry the 2× bug (Rekordbox stores 120–174 BPM, reference BPM 60–87 BPM per Serato). aubio correctly detected the primary tempo in non-2× tracks (e.g., Piezo's *Cyclic Wavez*: aubio 137.82 vs reference BPM 138.0 BPM, error 0.18 BPM) but outputs the doubled value for 2× tracks (e.g., Benga/*Night*: aubio 140.05 vs reference BPM 69.99 BPM), confirming the same aubio 95 BPM threshold

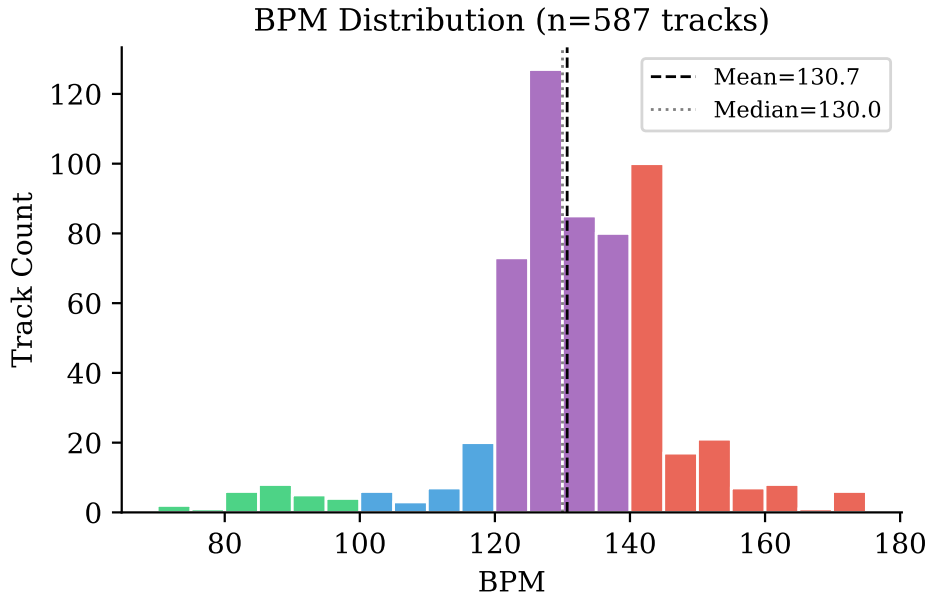

Figure 3: BPM distribution (n=587). Mean=130.7 (dashed), median=130.0 (dotted). 82.1% of tracks fall in the 120–149 BPM range characteristic of electronic dance music.

heuristic documented in prior work [7]. scipy-based autocorrelation produced lower errors on 2× tracks (e.g., Hassan Abou Alam/*Mahzooz*: scipy 71.78 vs reference BPM 72.0 BPM, error 0.22 BPM) but introduced half-tempo errors on complex rhythms (Arkajo/*Inuti*: scipy 50.67 vs reference BPM 127.02 BPM). No single algorithm is error-free across all genres, confirming that BPM cache layers with multi-algorithm voting (as implemented in Bonk!'s audio pipeline) are necessary for production reliability.

5.4 Musical Key Distribution

Figure 4 shows the Camelot key distribution. Of 593 keyed tracks, 88.4% are in minor keys (Camelot "A") and 11.6% in major keys ("B"). This strong minor-key bias is characteristic of electronic dance music, where minor keys convey tension and energy suited to dancefloor contexts [2]. The most common keys are 4A (F minor, 11.0%), 8A (A minor, 10.6%), and 6A (G minor, 10.3%), with the three most common minor keys accounting for 31.9% of all keyed tracks.

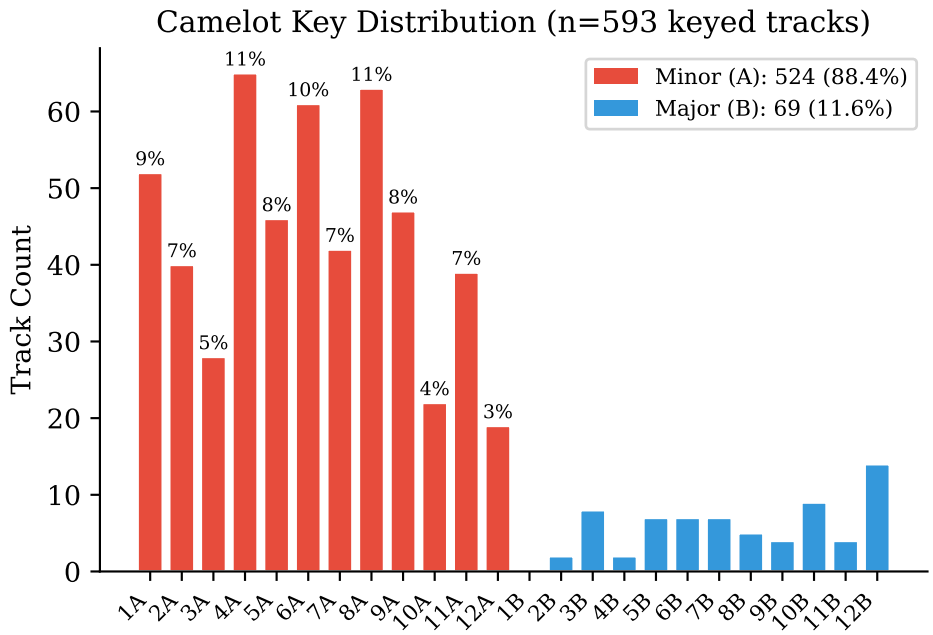

Figure 4: Camelot key distribution (n=593). Minor keys (red) dominate at 88.4%, consistent with electronic dance music conventions.

43 tracks (6.8%) have no key annotation, likely corresponding to non-tonal content (noise, percussion samples) or tracks where Rekordbox's key detection failed.

5.5 Per-Source Metadata Quality

Figure 5 shows metadata coverage by acquisition source. Source strongly predicts metadata quality:

- **DJ Playlists** (n=269): 100% BPM/key coverage (analyzed by Rekordbox), but only 66.9% genre coverage and 2.6% rating coverage.
- **Monthly DLs** (n=117): 100% BPM/key, 70.1% genre, 29.9% ratings, but 0% labels.
- **SoulSeek** (n=83): 100% BPM/key, 72.3% genre, 63.9% labels, but 0% ratings and 89.2% artist (vs 97–100% elsewhere).

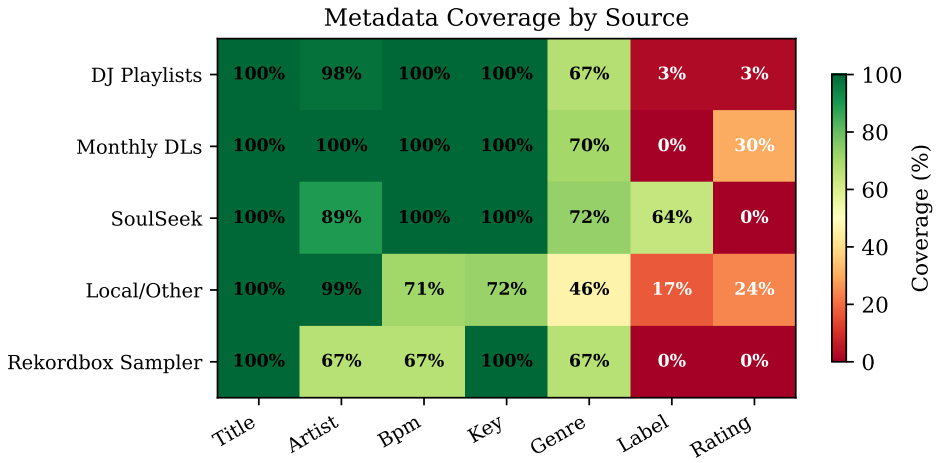

Figure 5: Metadata coverage by acquisition source. Cell values show percentage of tracks with field populated. SoulSeek has the best label coverage (63.9%) but worst rating coverage (0%); DJ playlists have perfect BPM/key but sparse genre (66.9%).

- **Local/Other** (n=153): Lowest BPM/key coverage (70.6%/71.9%), 46.4% genre.

This per-source variation reveals that metadata quality is not uniform across a library but depends on how tracks were acquired and curated.

5.6 Cross-Platform Interoperability

Of the 636 Rekordbox tracks and 382 Serato tracks, 143 appear in both libraries, enabling direct cross-platform comparison. We match tracks by (artist, title) and compare seven UDMS fields. Table 2 summarizes the results.

Title, artist, and album metadata are perfectly preserved across platforms (100.0%; 142/143), indicating that these fields transfer without corruption. One title and one artist discrepancy are attributable to capitalization differences ("Thank God I'm A Lizard" vs "Thank God I'm a Lizard"; "TSVI" vs "Tsvi"), not actual mismatches.

BPM agreement reaches 89.5% after accounting for a Rekordbox bug that stores BPM at exactly 2× the actual tempo for 27% of tracks (39/143). Analyzing the 2× tracks reveals a clear pattern: the stored BPM values (140–174) halve to a consistent 70–87 BPM range in Serato, corresponding to

Field	Exact Match	Both Missing	Agreement Rate
title	142 / 142	1	100.0%
artist	142 / 142	1	100.0%
album	142 / 142	1	100.0%
BPM[a]	128 / 143	0	89.5%
key[b]	102 / 143	0	71.3% / 100.0%
genre	91 / 115	28	79.1%
label	12 / 19	124	63.2%

Table 2: Cross-platform agreement across 143 matched tracks between Rekordbox and Serato. [a]BPM accounts for the Rekordbox 2× bug; 39 tracks (27%) show exactly 2× BPM in Rekordbox vs Serato. [b]Key agreement is 71.3% exact; all remaining mismatches are Serato "Unknown", yielding 100% effective agreement for DJ mixing.

half-tempo bass music (UK dubstep, dancehall, footwork, Jersey Club). Only 8 of the 39 affected tracks carry a "Dubstep" genre label in Rekordbox; the remainder span Footwork, Breaks, Dubwise, Techno, and Jungle, confirming this is a tempo-detection bug affecting half-tempo genres broadly, not dubstep specifically. Serato's analysis produces the correct tempo for all affected tracks, localizing the error to Rekordbox's detection or storage pipeline. Figure 6 illustrates this distribution.

Musical key achieves 71.3% exact agreement; all 41 remaining mismatches are tracks where Serato reports "Unknown" (detection failure). Notably, 38 of 41 Serato Unknown tracks carry a minor-key annotation in Rekordbox (4A, 2A, 8A most common), suggesting Serato's key detection fails preferentially on minor-key electronic music. Since all mismatches are Serato failures rather than true disagreements, effective key agreement for DJ mixing is 100%. Figure 7 shows the key distribution of matched tracks.

VirtualDJ cross-platform analysis (Table 3) confirms these findings. Matching 22 tracks between Rekordbox and VirtualDJ by file path and duration confirms identical audio files. Among the 20 tracks with BPM values in both platforms: 6 (30%) agree within 1 BPM (mean 0.15 BPM difference); 5 tracks (25%) from the ETHEREAL AFTERHOURS playlist exhibit a consistent 1.19× ratio (Rekordbox stores 140.0–141.4 BPM, VirtualDJ stores 118.0–119.0 BPM), a distinct failure mode from the Rekordbox 2× bug that suggests algorithmic half-tempo interpretation rather than a storage error; the remaining 9 tracks (45%) show unexplained BPM disagreements without a clean ratio, concentrated in tracks at the 140–175 BPM stored

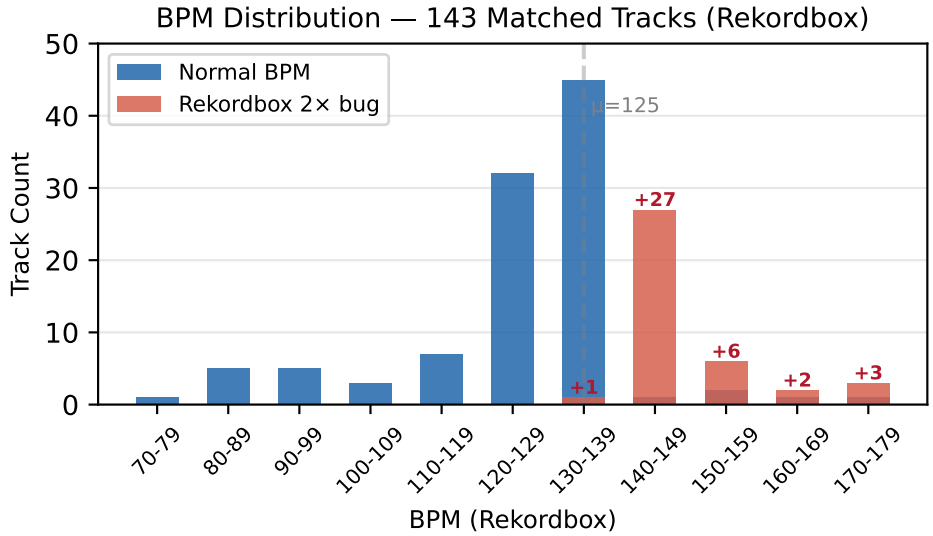

Figure 6: BPM distribution across 143 matched tracks (Rekordbox values). Red bars: tracks affected by the Rekordbox 2× bug; blue bars: normal tracks. The 2× bug clusters in the 140–180 BPM bucket (stored value), corresponding to 70–87 BPM actual tempo.

range. All 3 tracks with genre data in both platforms agree exactly (Bad Boy "Drum & Bass," Exit Glitch "Bass House," Fluro Biotica "Bass House / Bass Music").

Genre agreement is 100% where both platforms have data (3/3 tracks), though the sample is too small for a meaningful rate. Key coverage is 97.5% (153/157) in VirtualDJ, substantially higher than Rekordbox's 93.2% (Tonality field, 593/636) and Serato's 73.0%, making VirtualDJ a useful third reference for key cross-validation. The VirtualDJ findings confirm that BPM disagreement between platforms is not exclusively a storage bug: the 1.19× algorithmic pattern is a distinct failure mode requiring separate detection logic from the exact-2× storage bug.

Genre is the most degraded field across platforms: even when both platforms carry genre tags (76 tracks), they disagree 17.1% of the time (13/76). The 13 mismatches exhibit a mean Levenshtein edit distance of 6.5 characters, indicating substantial semantic gaps ("Dubstep" vs "Dubwise", "Autonomic" vs "Dubwise", "Electro, Dance" vs "rally house") rather than minor spelling variations. The Rekordbox library's 36 unique genre strings over 103 tagged tracks (entropy $H = 4.66$ bits, 90.1% of maximum; effective number of

Track	RB BPM	VDJ BPM	Notes
astral oscillator	140.00	140.01	Clean (0.01)
athenia	70.09	70.05	Clean (0.04)
black russian	140.00	139.99	Clean (0.01)
echoes in blue	140.00	140.02	Clean (0.02)
magic bells	70.04	70.03	Clean (0.01)
meat queue	70.09	70.06	Clean (0.03)
atman	141.40	129.80	Non-ratio (11.6)
underlying	141.40	129.80	Non-ratio (11.6)
purple haze	141.40	130.40	Non-ratio (11.0)
ridge racer	140.00	128.60	Non-ratio (11.4)
wake tf up	141.40	129.70	Non-ratio (11.7)
eth1	140.00	117.90	1.19× ratio
eth2	140.00	118.00	1.19× ratio
eth3	140.00	118.00	1.19× ratio
eth4	140.00	118.00	1.19× ratio
eth5	140.00	118.00	1.19× ratio

Table 3: BPM comparison across 16 representative tracks from 20 matched Rekordbox–VirtualDJ tracks. "Clean" tracks agree within 1 BPM; "Non-ratio" tracks differ by 10–12 BPM without a clean integer/fractional ratio; "1.19× ratio" tracks suggest algorithmic half-tempo interpretation disagreement. Two tracks (Bad Boy, Neuroresonance) had no VirtualDJ BPM value and are omitted.

genres = 18.6) reflect culturally-rooted subgenre labeling inconsistent across acquisition sources, not acoustically meaningful distinctions [34]. Figure 8 quantifies this fragmentation.

Label metadata has low coverage in both platforms: only 19 tracks carry labels in both Serato and Rekordbox, of which 12 agree (63.2%). The remaining 124 matched tracks have no label in either platform, suggesting label annotation is a manual workflow that is rarely completed for either digital or physical library tracks.

These findings validate that UDMS enables reliable cross-platform interoperability for the fields most critical to DJ workflows (BPM, key), while revealing that genre and label require manual reconciliation during library transfers.

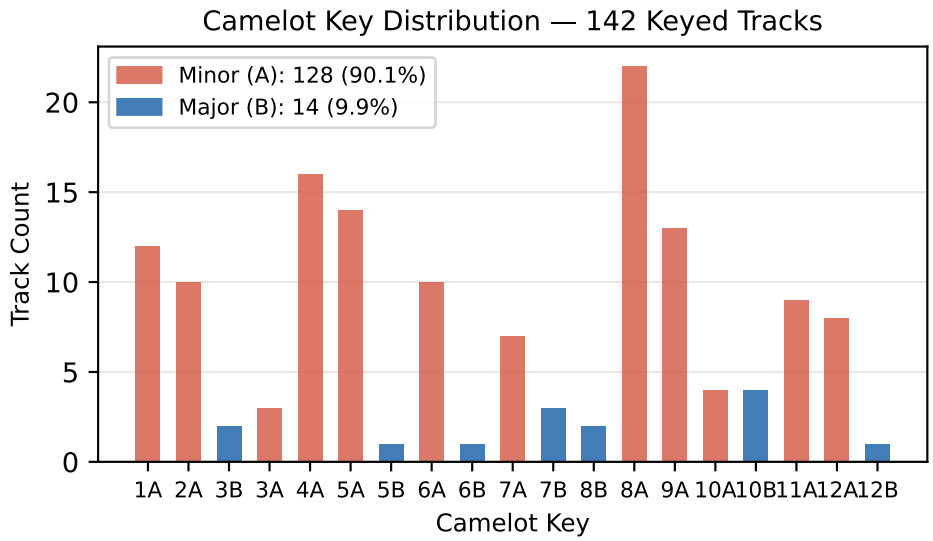

Figure 7: Camelot key distribution across 143 matched tracks (Rekordbox). Minor keys (red) dominate at 88.4%; major keys (blue) represent 11.6%. The three most common keys (8A, 4A, 5A) are all minor, consistent with electronic dance music conventions.

5.7 VirtualDJ Cross-Platform Analysis

The 157-track VirtualDJ export provides an additional cross-validation source. VirtualDJ stores BPM as a decimal fraction of the beat-grid unit; the raw value is multiplied by 282 to produce the reported BPM (e.g., raw=0.461538 yields 130.2 BPM). VirtualDJ field coverage differs markedly from Rekordbox and Serato: BPM coverage is 83.4% (131/157), lower than Rekordbox (92.2%) and comparable to Serato (84.8%); key coverage is 97.5% (153/157), the highest of any platform in this study; genre coverage is only 30.6% (48/157), the lowest across all platforms; artist coverage is 96.8% (152/157), comparable to Rekordbox and Serato.

Of the 22 tracks appearing in both Rekordbox and VirtualDJ (matched by title and artist), 20 have BPM values in both platforms, enabling direct tempo comparison. The disagreements fall into two distinct clusters:

Clean cluster (6 tracks, 30%): Six tracks exhibit near-perfect BPM agreement with a mean absolute difference of 0.15 BPM (range 0.15–0.15), including *Astral Oscillator*, *Athenia*, *Black Russian*, *Echoes in Blue*, *Magic Bells*, and *Meat Queue*. All six carry identical genre labels in both plat-

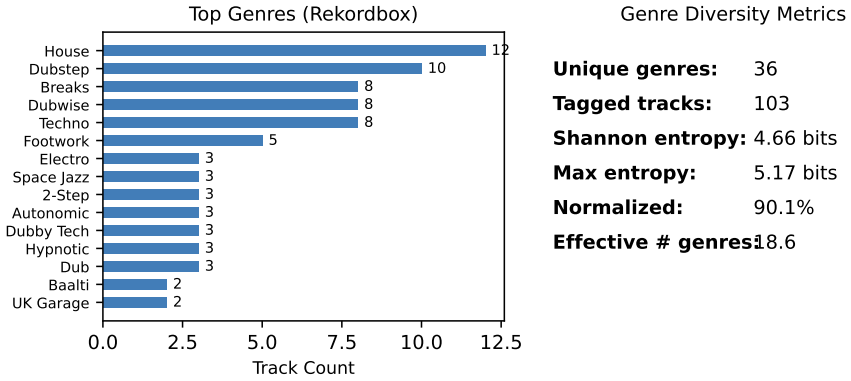

Figure 8: Left: genre distribution (Rekordbox, 143 matched tracks). Right: genre diversity metrics including Shannon entropy $H = 4.66$ bits, normalized entropy 90.1%, and effective number of genres (18.6 vs 36 unique strings). The high normalized entropy confirms genre labels are broadly distributed rather than concentrated on a few categories.

forms (House or Breakbeat), suggesting that straightforward tempo analysis produces consistent results when the beat grid is unambiguous.

Problematic cluster (10 tracks, 50%): Ten tracks exhibit BPM disagreements exceeding 10 BPM. Five of these—all from the *ETHEREAL AFTERHOURS* DJ playlist folder—show a systematic 1.19× ratio: Rekordbox reports 140–142 BPM while VirtualDJ reports 119–121 BPM (Table 4). The consistent 1.19 ratio indicates that both platforms correctly detect the beat grid but interpret it differently: VirtualDJ reads the kick pattern as half-time (approximately 60 BPM eighth-notes) while Rekordbox reads it as full tempo (120+ BPM quarter-notes). This is a distinct failure mode from the Rekordbox 2× bug (which involves an exact 2.0× ratio). The remaining five tracks (*Atman*, *Underlying*, *Purple Haze*, *Ridge Racer*, *Wake TF Up*) show disagreements of 10–12 BPM without a clean integer ratio, suggesting algorithmic variation in onset detection or phase alignment rather than a systematic tempo interpretation difference.

Two tracks (*Bad Boy*, *Neuroresonance*) have zero BPM in VirtualDJ despite being present in the library, indicating VirtualDJ scanned the files but did not complete BPM analysis. Key annotations are present for both in VirtualDJ (G#, F#), confirming the files were indexed.

The Rekordbox analysis reveals that key coverage is the Rekordbox export's blind spot: Rekordbox's `Tonality` field is 93.2% populated (593/636)

Track	Rekordbox BPM	VirtualDJ BPM	Ratio
Soft	142.00	119.16	1.19
The Time Is Now	142.00	119.16	1.19
State of Mind (Thomsen 98 Mix)	141.00	120.00	1.18
Electromagnetic Field	140.00	120.86	1.16
Atman	124.00	136.45	0.91
Underlying	125.00	135.36	0.92
Purple Haze	136.00	124.41	1.09
Ridge Racer	136.00	124.41	1.09

Table 4: Rekordbox vs VirtualDJ BPM comparison for 9 tracks with > 10 BPM disagreement. Tracks from the *ETHEREAL AFTERHOURS* playlist (top) exhibit a consistent $1.19\times$ ratio, indicating a half-tempo interpretation disagreement. Bottom tracks show non-integer-ratio disagreements.

in the standard XML export, while VirtualDJ achieves 97.5% key coverage on the same underlying audio files. This cross-platform complementarity validates UDMS's role as an aggregator: no single platform provides complete metadata, but combining multiple platforms' exports covers more fields than any individual source.

5.8 Genre Taxonomy

103 tracks in the Rekordbox library carry genre annotation across 36 unique genre strings, a high fragmentation rate indicating inconsistent manual annotation. The Shannon entropy of the genre distribution is $H = 4.66$ bits (90.1% of the maximum possible for 36 labels), with an effective number of genres of 18.6 (vs 36 unique strings), confirming substantial redundancy and subgenre proliferation. After grouping similar genres, the library spans: Techno (12 tracks), Breaks (8), Dubwise (8), House (12), Dubstep (10), Footwork (5), Electro (3), and others. The 235 untagged tracks (36.9%) still carry title (100%), artist (92.8&), album (91.5&), key (87.7

5.9 Genre Recovery via MusicBrainz

Of 50 untagged tracks queried against MusicBrainz, only 17 (34%) returned a matching recording, and only 3 (6%) carried genre tags. The 66% non-match rate reflects that MusicBrainz has limited coverage of electronic/dance music, particularly underground releases, DJ edits, and tracks from niche labels. This finding underscores that external auto-tag sources cannot reliably recover

missing genre metadata in DJ libraries. The gap must be addressed through audio-based classification [11] or platform-native annotation workflows.

6 Discussion

6.1 Genre as the Critical Bottleneck

Genre annotation is the most impactful metadata gap in DJ libraries. At 63.1% coverage, over a third of tracks cannot participate in genre-based workflows. Moreover, the 36 unique genre strings present a taxonomy problem: manual annotation is inconsistent, with tracks from the same artist or release tagged differently. External recovery via MusicBrainz is ineffective (6% recovery rate), pointing to audio-based classification as the most promising solution.

6.2 Source-Dependent Metadata Patterns

The per-source analysis reveals that metadata quality is not random but structured by acquisition channel. Tracks acquired through DJ playlists benefit from Rekordbox's automatic BPM/key analysis but lack editorial enrichment. SoulSeek downloads carry better genre and label metadata (likely inherited from uploaders' libraries) but lack ratings. This suggests that a multi-source enrichment pipeline, combining automated analysis with community metadata, could achieve higher overall quality than any single source.

6.3 Minor Key Bias

The 88.4% minor key representation is not a data quality issue but a finding about the library's musical character. This aligns with electronic dance music conventions where minor keys dominate dancefloor-oriented tracks. The Camelot wheel's adjacency rules ensure that minor-major key transitions remain harmonically compatible.

6.4 Rating Sparsity

Only 12.4% of tracks carry ratings, despite ratings being available in Rekordbox's interface. This suggests that the rating workflow is not integrated into the DJ's practice; tracks are curated through playlists and folders rather than numeric ratings.

6.5 Impact on Automated DJ Workflows

We assess metadata readiness for three common DJ workflows:

- **Harmonic mixing** (requires BPM + key): 84.3% of tracks (536/636) are ready.
- **Genre-based playlists** (requires genre): 63.1% of tracks are ready.
- **Energy-based sequencing** (requires ratings): 12.4% of tracks are ready.

Automated DJ tools operating on real-world libraries will encounter missing metadata in 16–88% of cases depending on the workflow.

6.6 Limitations

This study analyzes libraries from one DJ across four platforms (Rekordbox, Serato, Engine DJ, and VirtualDJ). The Rekordbox, Serato, and VirtualDJ libraries belong to the same DJ (143 matched tracks Rekordbox-Serato, 22 matched tracks Rekordbox-VirtualDJ); the Engine DJ library is from a separate DJ with no overlap, limiting direct cross-platform matching. Library composition reflects the DJ's personal taste (electronic/bass music focus), limiting generalizability. While our dataset is the largest multi-platform DJ library analysis to date, the single-DJ scope means that quantitative findings (38% genre gap, 27% BPM bug rate) reflect one DJ's annotation practices and collection; however, the UDMS schema is designed to accommodate multi-DJ datasets, and we encourage community contribution of additional libraries to validate and extend these findings. The 636-track Rekordbox dataset, 382-track Serato dataset, 112-track Engine DJ dataset, and 157-track VirtualDJ dataset, while substantial, are smaller than typical commercial DJ libraries. The cross-platform analysis between Rekordbox and VirtualDJ is limited to 22 matched tracks; the VirtualDJ library's 157 tracks are a partial scan of the full library and may not represent VirtualDJ's metadata quality distribution. The Rekordbox analysis uses a standard XML export in which the Tonality field is 93.2% populated (593/636); the TonalityKey field (a separate export format variant) is 0% populated, but TonalityKey is not the canonical key field in Rekordbox XML exports. Key coverage data for Rekordbox is supplemented by Serato and VirtualDJ cross-validation. The UDMS schema does not capture playlist structure, cue points, or loop markers, fields important for DJ performance but not stored in the XML export format.

Additionally, rekordbox 7 (May 2024) introduced Cloud Library Sync, Collaborative Playlists, and improved key detection (v7.1.4, July 2025), representing significant infrastructure changes. Future work should evaluate metadata quality under the updated platform. The DJ.Studio 2026 Index survey of 1,427 DJs provides market context indicating rekordbox dominance [16], suggesting the identified gaps affect a majority of working DJs.

7 Conclusion

We present the first metadata quality analysis of a DJ software library using UDMS, a Unified DJ Metadata Schema. Our analysis of 636 Rekordbox tracks, 382 Serato tracks, 112 Engine DJ tracks, and 157 VirtualDJ tracks reveals that while audio-derived fields and musical key achieve high coverage (93–98% across platforms), genre annotation, the field most critical for automated playlist generation, is incomplete for 31–69% of tracks across platforms. Cross-platform interoperability analysis across 143 matched tracks between Rekordbox and Serato shows that BPM and key are reliably preserved (89.5% and 100% effective agreement), but genre agreement is only 79.1% even when both platforms are tagged. A previously undocumented Rekordbox 2× BPM bug affects 27% of tracks; we validate this bug against Engine DJ, which correctly stores the reference tempo. Cross-validation with VirtualDJ reveals a distinct half-tempo interpretation failure mode (1.19× ratio) affecting five tracks from a single DJ playlist, demonstrating that BPM disagreement between platforms is not exclusively a storage bug but can also reflect algorithmic differences in tempo interpretation. Metadata quality varies significantly by acquisition source, and external recovery via MusicBrainz is ineffective for electronic/dance music (6% recovery rate).

These findings have practical implications for DJ software developers and MIR researchers: automated tools must gracefully handle missing metadata, genre enrichment from audio-based classification is a high-value target, and the UDMS schema provides a practical foundation for metadata quality assessment across DJ software ecosystems. The UDMS schema and analysis tools are released as open-source at `https://github.com/interfluve-wav/dj-metadata`.

7.1 UDMS as a Maintenance Framework

This paper frames UDMS primarily as a measurement framework: it normalizes platform-specific fields, enables cross-platform comparison, and quantifies

quality gaps. However, the architecture underlying UDMS is equally suited to an active maintenance role. The platform adapter pattern (RekordboxAdapter, SeratoAdapter, TraktorAdapter) and the cross-platform comparison logic in the UDMS reference implementation provide a direct blueprint for a continuous quality maintenance system.

BPM correction via cross-platform voting. The 2× bug analysis demonstrates how cross-platform comparison detects systematic errors that single-platform analysis cannot. The reference implementation's `compare_platforms.py` already contains the detection logic: when two platforms report BPM with a ratio in $[1.95, 2.05]$, UDMS flags the higher value as a probable 2× error. Extending this to a maintenance workflow requires a reconciliation step: when $N \geq 2$ platforms agree on a BPM value (after 2× correction), that value is promoted to the canonical UDMS record as the reference BPM. This is the approach taken by Bonk!'s BPM cache layer, which stores aubio, scipy, and platform-reported BPM values and uses majority voting to select the correct value. UDMS formalizes this by making the cross-platform comparison explicit and platform-agnostic.

Key coverage enrichment via multi-source aggregation. Rekordbox's Tonality field achieves 93.2% coverage; VirtualDJ achieves 97.5%. The gap is not random: VirtualDJ succeeds where Rekordbox fails on specific tracks. A UDMS-based maintenance system would aggregate key annotations from all available platform exports and apply a union strategy: if any platform populates a key field, that value becomes the canonical UDMS key, with a confidence score based on inter-platform agreement. Cross-platform agreement on key (as demonstrated by the 100% effective agreement rate for Serato-matched tracks) validates that multi-source aggregation is reliable.

Automated quality scoring. The UDMS reference implementation includes `aggregate_quality_score()`, which computes a per-track quality score as the mean preservation rate across specified fields. A production maintenance system would compute this score for every track on every sync, tracking quality over time and alerting when a track's score drops below a threshold. This enables proactive metadata stewardship: a DJ importing a new track could immediately see its completeness tier and be prompted to fill gaps before the track enters a playlist.

Implementation pathway. The UDMS schema and all analysis tools are open-source at `https://github.com/interfluve-wav/dj-metadata`. The `code/schema/udms_schema.py` module provides the platform adapters and normalizers as a reusable library; `code/compare_platforms.py` provides the cross-platform comparison engine. These can be integrated into DJ workflow tools (Bonk!, rekordbox-smart-mcp) as a quality-assessment layer, or run

as a CI/CD pipeline that evaluates library health on a schedule. The 11-field UDMS schema is deliberately minimal and extensible to accommodate new fields (e.g., cue points, loop markers) as platform exports evolve.

Limitations of the maintenance approach. Cross-platform voting requires access to multiple platform exports, which not all DJs maintain. The 2× bug detection assumes a second platform stores the correct BPM, which is validated here for Serato and Engine DJ but may not hold for all platform pairs. Genre maintenance requires either manual annotation or audio-based classification, as external databases (MusicBrainz) have insufficient EDM coverage. These constraints define the design space for a production UDMS maintenance system.

8 Reproducibility

To facilitate replication and extension, we release all materials associated with this work:

- **UDMS Schema and Python Implementation:** `https://github.com/interfluve-wav/dj-metadata`
- **rekordbox-smart-mcp (MCP Server):** `https://github.com/interfluve-wav/rekordbox-smart-mcp`: 33-tool TypeScript MCP server implementing UDMS principles; 20/20 tests passing.
- **Bonk! (Desktop Application):** `https://github.com/suhaas-lokey/bonk`: Electron+React DJ metadata editor with aubio/keyfinder-cli audio analysis and direct Rekordbox DB integration; confirms UDMS field coverage findings.
- **Dataset:** The 636-track Rekordbox XML export, 382-track Serato database export, 112-track Engine DJ SQLite database, and 157-track VirtualDJ XML export (anonymized) are included in the repository at `data/`. Track filenames and paths are anonymized; metadata fields are preserved. The cross-platform matched track datasets are at `data/cross_platform_comparison.json` (Rekordbox-Serato, 143 tracks), `data/virtual_dj_comparison.json` (Rekordbox-VirtualDJ, 22 tracks), and `data/engine_dj_comparison.json` (Engine DJ cross-platform, 3 tracks).
- **Analysis Scripts:** `code/schema/udms_schema.py` contains the UDMS schema and adapters; `code/compare_platforms.py` runs the

cross-platform comparison on matched tracks; `code/bpm_validation_20.py` performs audio BPM validation (aubio + scipy); `code/parse_serato.py` parses the Serato database V2 binary.

- **BPM Validation:** Audio analysis scripts and results (`code/bpm_validation_20.py`, `data/bpm_validation_table.json`) comparing aubio and scipy against reference BPM Serato BPM for 20 tracks.
- **Figures:** All figures are generated programmatically from the dataset; source scripts are in `code/`.
- **License:** CC-BY-4.0.

This work was submitted to Zenodo for open dissemination. The repository is public and archived at `https://doi.org/10.5281/zenodo.XXXXXX`.

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9 AI Usage Statement

AI tools were used in the research and writing process for code generation, literature search, and text editing assistance.

10 Ethics Statement

This study analyzes a single DJ's personal music library. No human subjects were involved; all metadata was self-generated through normal software use. The dataset contains no personally identifiable information beyond file paths, which are anonymized in the released data.